

Accessibility of Formal Theorem Proving via LLM Agents:

A Preliminary Report for a PhD application in Innovation Research

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1 Introduction and Research Question

The study addresses one question: “Can a researcher with no prior knowledge of the Lean theorem prover, but with standard computational literacy (e.g., an ecologist proficient in R), produce a formal mathematical proof using LLM agents?” The motivation is the democratising potential of LLMs in formal verification. The workflow centred on Aristotle, which autonomously generates Lean environments and files; Claude and Gemini were used as intermediate models to iteratively craft, test and refine the natural-language prompts.

2 Key Findings

Algorithmic error detection in source material. Integrating Lean exposed errors in the foundational assumptions. A pre-existing complete proof (Q.E.D) was found to attach the origin to Quadrant I rather than treating it as isolated, a useful demonstration of the agent’s value for rigorous premise checking, even on results already considered settled.

The complexity-inversion phenomenon. Initial intuitions about which sub-proofs would be hard turned out to be highly inaccurate.

- *Unexpected bottlenecks:* “common-knowledge” results were often the hardest to formalise. Establishing that the density of lattice points visible from the origin is $6/\pi^2$ required Aristotle to devise its own proof strategy.
- *Unexpected efficiencies:* domains that looked structurally daunting (e.g., D-algebra proofs) were dispatched almost instantly. The transition from planning to implementation in formal verification is dominated by “unknown unknowns.”

Conceptual roadblocks and multi-model mediation. Foundational logic errors repeatedly stalled progress (e.g., misapplied concepts, the unnecessary invocation of cardinality). Recovery typically required stepping back and re-using Claude and Gemini to reformulate the underlying theorems.

Human versus machine cognition

In the researcher’s mind, mathematical concepts exist as an implicit, cohesive cloud of ideas. For an ecologist, this intuitive cloud is shaped by a diverse blend of empirical metrics and structural abstractions: measuring diameter at breast height (DBH); modeling trees as geometric cones or cylinders; applying Bergmann’s Rule; evaluating canopy surfaces via the Leaf Area Index;

considering fractal shapes; or estimating spatial distributions and diversity with probabilistic models (e.g., the Shannon or Simpson indices). Formalising this multi-faceted intuition, and finding the precise, rigid vocabulary the machine requires (e.g., distinguishing geometry from topology), remains a dominant cognitive barrier.

The semantic-alignment gap and cascading drift. The architecture reliably produced syntactically valid Lean (“a proof”), but a critical epistemic question remains: did it prove the theorem the researcher intended? This is exacerbated by the multi-agent setup. Each translation step (human intuition → Claude → Gemini → Aristotle’s formal logic) might introduce a small margin of error, and these deviations might compound along the chain, a telephone-game failure mode in which syntactically perfect Lean encodes a subtly distorted version of the original premise.

Asymptotic progression. Effort followed a sharp Pareto-like distribution, with steeply diminishing marginal returns: driving the proof from 0% to 80% was efficient, while each additional percentage point toward a fully verified proof (still open) demanded disproportionately more time and compute. For the user, each additional percentage point demands hours of iterative prompting, with attendant frustration and dwindling returns on attention. For the provider (e.g. Anthropic, Google, Harmonic), every iteration consumes inference compute and electricity that scale with effort expended, not with verified output. The long tail is therefore simultaneously the user’s bottleneck and the provider’s least economically viable segment, thus relevant to pricing models and to the sustainability of subsidised access (in the currently free Aristotle).

3 Methodological Limitations and Potential Biases

The problem of induction ($n = 1$). As a bounded three-day study with a single user, the findings are a proof of concept and cannot support general usability claims for an “average user.”

Autoethnographic bias. Acting as the primary test subject is a notable limitation. Justification: The prompter has an underlying familiarity with the mathematical content being formalised. A target user without this foundation (e.g., the arbitrary ecologist) may lack the intuition currently required to steer both proof strategy and LLM prompting.

4 Open Questions and Future Work

- *Absolute verification:* whether 100 % formal correctness can be reached reliably and consistently by a non-expert through a multi-agent stack (Aristotle guided by Claude/Gemini) remains open.
- *Expert peer review:* the generated Lean proofs, and the report’s broader claims, require evaluation by the formal-mathematics and Lean communities.
- *Value of partial results:* it is unclear whether the partial artefacts produced have standalone value for other researchers or warrant inclusion into the wider Lean ecosystem (e.g., `mathlib`).